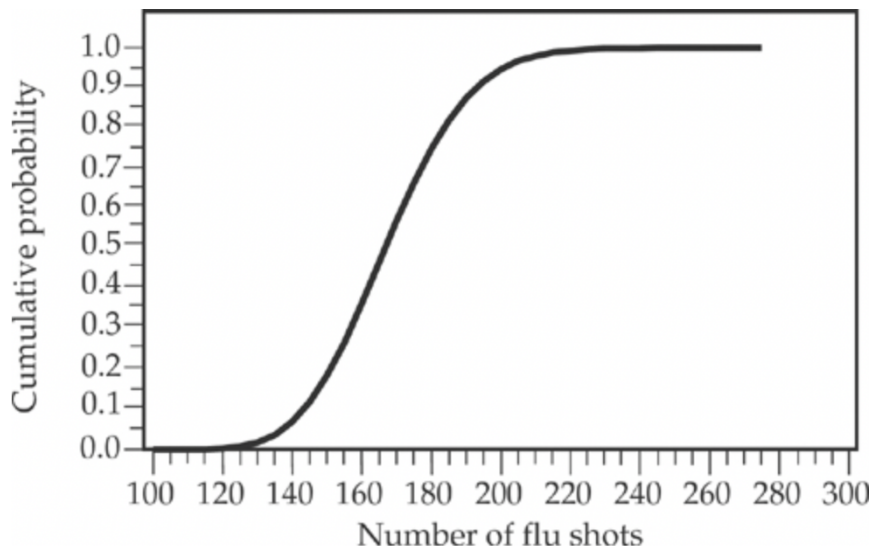


1. An outlier is an observation that

- ☐ A. is seen more frequently than the other observations in the data set
- ☐ B. is seen less frequently than the other observations in the data set
- ☐ C. is always smaller than the other observations in the data set
- ☐ D. is always larger than the other observations in the data set
- ☐ E. is significantly different from the other observations in the data set

2. During flu season, a city medical center needs to keep a large supply of flu shots. A nurse's aid compiles data on the number of flu shots given per day in the past few years during flu season. A cumulative probability chart of the collected data is as follows:



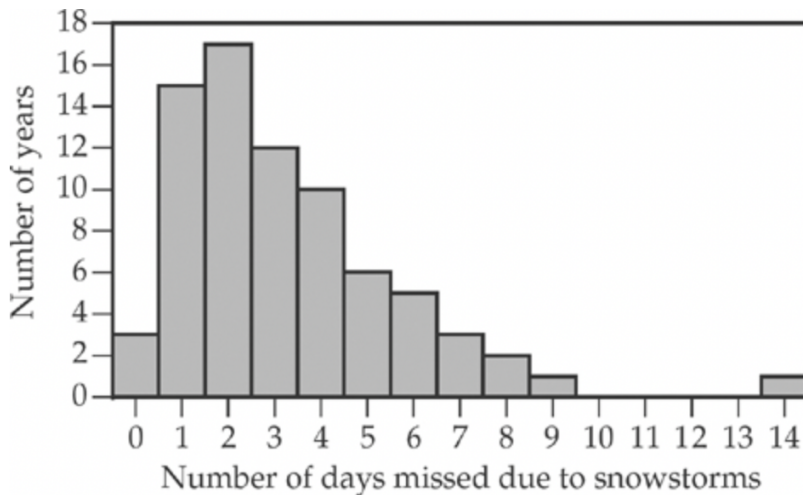
How many flu shots should the center store every day to meet the demand on 95 percent of the days?

- ☐ A. At most 190
- ☐ B. At most 140
- ☐ C. Exactly 170
- ☐ D. At least 150
- ☐ E. At least 200

3. A large company has offices in two locations, one in New Jersey and one in Utah. The mean salary of office assistants in the New Jersey office is \$28,500. The mean salary of office assistants in the Utah office is \$22,500. The New Jersey office has 128 office assistants and the Utah office has 32 office assistants. What is the mean salary paid to the office assistants in this company?

- ☐ A. \$22,500
- ☐ B. \$23,700
- ☐ C. \$25,500
- ☐ D. \$27,300
- ☐ E. \$28,500

4. In the northern United States, schools are sometimes closed during winter due to severe snowstorms. At the end of the school year, schools have to make up for the days missed. The following graph shows the frequency distribution of the number of days missed due to snowstorms per year using data collected from the past 75 years.



Which of the following should be used to describe the center of this distribution?

- ☐ A. Mean, because it is an unbiased estimator
- ☐ B. Median, because the distribution is skewed
- ☐ C. *IQR*, because it excludes outliers and includes only the middle 50 percent of data
- ☐ D. First quartile, because the distribution is left skewed
- ☐ E. Standard deviation, because it is unaffected by the outliers

5. The probability that there will be an accident on Highway 48 each day depends on the weather. If the weather is dry that day, there is a 0.2% chance of an accident on Highway 48; if the weather is wet that day, there is a 1.0% chance of an accident. Today, the weather station announced that there is a 20% chance of the weather being wet. What is the probability that there will be an accident on Highway 48 today?

- ☐ A. 0.0004
- ☐ B. 0.0016
- ☐ C. 0.0020
- ☐ D. 0.0036
- ☐ E. 0.0060

6. An employment placement agency specializes in placing workers in jobs suited for them. From past experience, the agency knows that 20% of all the workers it places will no longer be at the position in which they were placed after one year; however, only 5% of those remaining after the first year leave during the next year. At the start of a year an employer hires 100 workers using this agency, then at the start of the next year the employer hires 100 more. How many of these 200 workers are expected to be on the job at the end of the second year?

- ☐ A. 140
- ☐ B. 144
- ☐ C. 152
- ☐ D. 156
- ☐ E. 171

7. The average number of calories in Yum-Yum Good candy bars is 210, with a standard deviation of 10. If the number of calories per candy bar is normally distributed, what percent of candy bars contain more than 225 calories?

- ☐ A. 66.8%
- ☐ B. 47.7%
- ☐ C. 43.3%
- ☐ D. 6.68%
- ☐ E. 3.34%

8. A publisher used standard boxes for shipping books. The mean weight of books packed per box is 25 pounds, with a standard deviation of two pounds. The mean weight of the boxes is one pound, with a standard deviation of 0.15 pounds. The mean weight of the packing material used per box is two pounds, with a standard deviation of 0.25 pounds. What is the standard deviation of the weights of the packed boxes?

- ☐ A. 28.000 pounds
- ☐ B. 5.290 pounds
- ☐ C. 4.085 pounds
- ☐ D. 2.400 pounds
- ☐ E. 2.021 pounds

9. The principal of a school is interested in estimating the average income per family of her students. She selects a random sample of students and collects information about their family income. A 95 percent confidence interval computed from this data for the mean income per family is (\$35,095, \$45,005). Which of the following provides the best interpretation of this confidence interval?

- ☐ A. 95 percent of the students in her school are from families whose income is between \$35,095 and \$45,005.
- ☐ B. There is a 95% probability that the families of all the students in this school have an income of between \$35,095 and \$45,005.
- ☐ C. If we were to take another sample of the same size and compute a 95 percent confidence interval, we would have a 95% chance of getting the interval (\$35,095, \$45,005).
- ☐ D. There is a 95% probability that the mean of another sample with the same size will fall between \$35,095 and \$45,005.
- ☐ E. There is a 95% probability that the mean income per family in the school is between \$35,095 and \$45,005.

10. The Department of Health plans to test the lead level in a specific park. Because a high lead level is harmful to children, the park will be closed if the lead level exceeds the allowed limit. The department randomly selects several locations in the park, gets soil samples from those locations, and tests the samples for their lead levels. Which of the following decisions would result from the type I error?

- ☐ A. Closing the park when the lead levels are within the allowed limit
- ☐ B. Keeping the park open when the lead levels are in excess of the allowed limit
- ☐ C. Closing the park when the lead levels are in excess of the allowed limit
- ☐ D. Keeping the park open when the lead levels are within the allowed limit
- ☐ E. Closing the park because of the increased noise level in the neighborhood